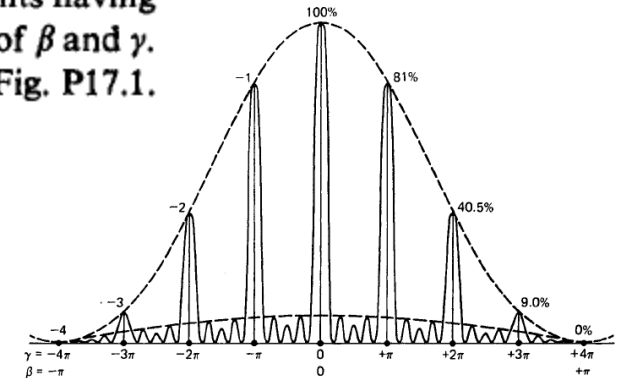


Practice Problems 11

Make a qualitative sketch for the intensity pattern for five equally spaced slits having $d/b = 4$. Label several points on the x axis with the corresponding values of β and γ .

Ans. See Fig. P17.1.



Nine coherent sources of microwaves in phase and with a wavelength of 2.50 cm are placed side by side in a straight line, 10.0 cm between centers. Compute (a) the angular width of the central maximum. Find the angular separation of (b) the principal maxima and (c) the subsidiary maxima.

Light of two wavelengths, $\lambda = 5600 \text{ \AA}$ and $\lambda = 5650 \text{ \AA}$, fall normally on a plane transmission grating having 2500 lines per centimeter. The emerging parallel light is focused on a flat screen by a lens of 120 cm focal length. Find the distance on the screen in centimeters between the two spectrum lines (a) in the first order and (b) in the second order.

A diffraction grating containing 5000 lines per centimeter is illuminated at various angles of incidence by light of wavelength $4000 \text{ \AA}$. Draw a graph of the deviation of the first-order diffraction beam from the direction of the incident light, using the angle of incidence from 0 to 90° plotted on the x axis.